
Space Weather Technical Forecast

Issued on Wednesday, 08 May 2024 at 13:27 Local

This technical forecast provides a four day assessment of space weather events. The probabilities stated below are for reaching or exceeding the given levels. For more information about space weather impacts please see the Met Office Space Weather Scales <https://www.metoffice.gov.uk/weather/learn-about/space-weather/uk-scales>

Space Weather Forecast Headline: Chance of further X-class flares (R3 - Strong radio blackouts) throughout. Chance of G2-G3/Moderate-Strong geomagnetic storm conditions on 10 May.

Analysis of Space Weather Activity over past 24 hours

Solar Activity: Solar activity is High with seven sunspots on the visible disc. AR3663 in the northwest and AR3664 in the southwest have both produced frequent M-class flares during the period, as well as each an X1.0 flare; at 08/0141 UTC from AR3663 and 08/0509 UTC from AR3664. The former is approaching the western limb, making analysis of its magnetic structure difficult. It appears to retain a delta configuration, although has seen a recent slight decline in penumbral extent. AR3664 remains by far the largest region on the disc and comprises a number of strong spots along with several delta spot configurations. The region lies in close proximity to AR3668 to the east, with which it is magnetically connected. The remaining sunspots on the disc are smaller, stable and magnetically simple.

A 'full-halo' CME associated with the LDE X1.0 flare of AR3664 at 08/0509 UTC is still being analysed, but initial output suggests an arrival time late on Day 3 (10 May).

Solar Wind / Geomagnetic Activity: Solar winds observed by ACE at L1 declined from elevated to slightly elevated levels through the period as the connection to the HSS of CH35/+ continued to ease. Speeds peaked at 555km/s at 07/1713 UTC. Density remained below average through the period. The Interplanetary Magnetic Field, Bt, persisted at weak levels with a peak of around 6nT at 07/1253 UTC. The north-south component, Bz, varied weakly between +5/-4nT. Phi angle was mostly positive (away from the sun). Geomagnetic activity was Quiet to Unsettled (Kp 1-3).

Energetic Particles / Solar Radiation: High energy proton flux (greater than 10MeV) observed at GOES18 was at normal background levels, although indicated a minor response following the X1.0 flare of AR3664 at 08/0509 UTC. High energy electron flux (greater than 2MeV) observed at GOES16 remained at largely background levels. The associated 24-hour fluence was well below the Active threshold. Observed fluence at 08/0001 UTC was 1.67e6 integrated pfu on a steady trend.

Four-Day Space Weather Forecast Summary

Solar Activity: High solar activity is expected to continue, with frequent M-class flares and a chance of isolated X-class flares throughout, principally from AR3663 and AR3664.

Solar Wind / Geomagnetic Activity: There are several potential Earth-directed CMEs in the forecast period. There is a slight chance for a couple of glancing impacts early on Day 2 (09 May) and also on Day 3 (10 May). A more significant CME is expected to arrive Day 3 (10 May), resulting from the X1.0 flare of AR3664 that occurred early on 08 May. Earth is under the influence of a slowly declining HSS (High speed stream) from CH35/+ with slightly elevated solar winds expected to fall to background levels by Day 3 (10 May). No further coronal hole enhancements are expected during the period.

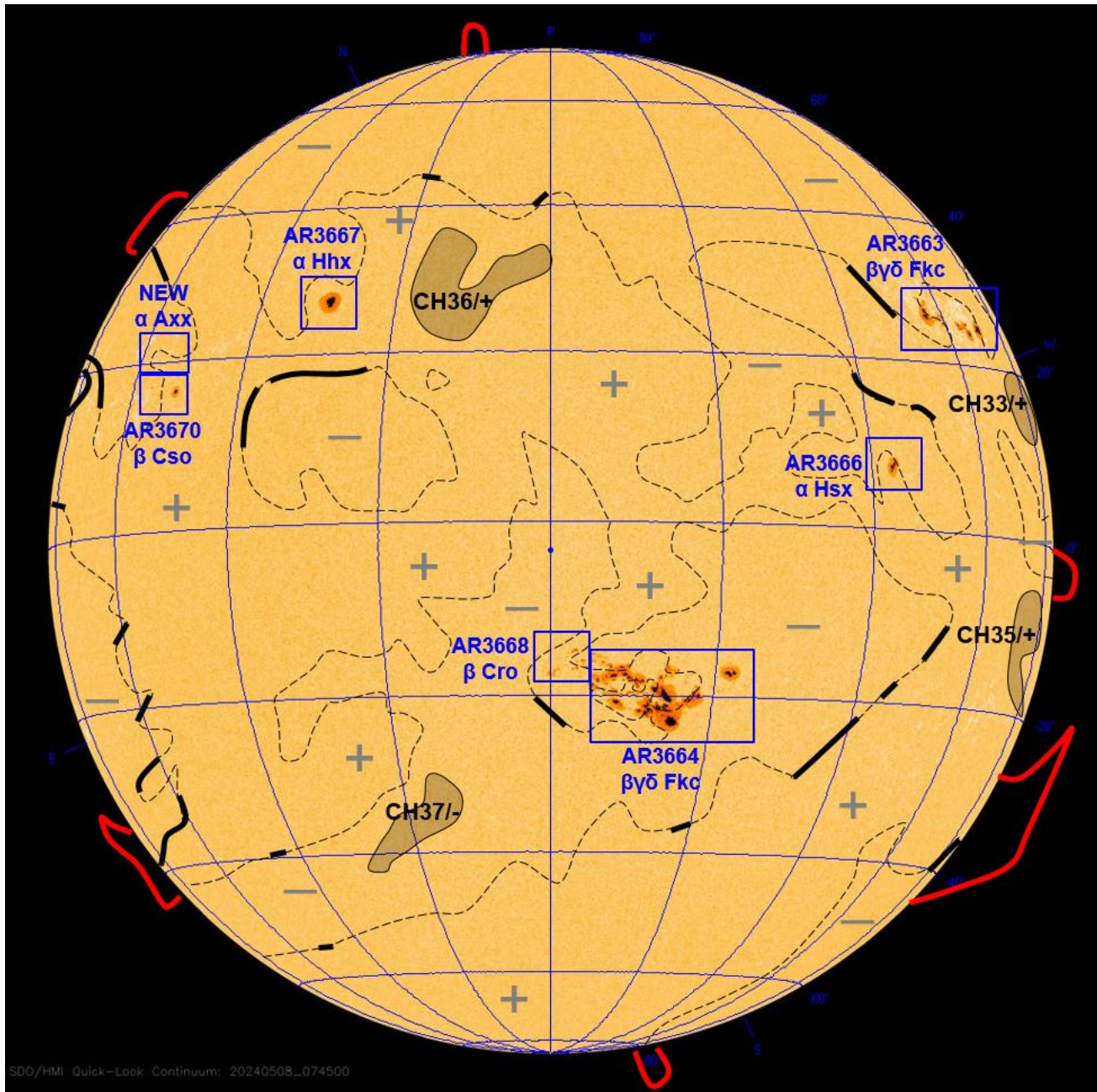
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Geomagnetic activity probably mostly remaining Quiet to Unsettled through the early guidance period, with a slight chance of Minor Storm (Kp5) intervals on Days 1 and 2 (08 and 09 May). Pending the forecast 'full-halo' CME arrival on Day 3 (10 May), activity should increase to G1-G2/Minor-Moderate Storm levels and a chance of peak G3/Strong Storm conditions, with activity projected to decline into Day 4 (11 May).

Energetic Particles / Solar Radiation: High energy proton flux is likely to remain at background levels through the period. There is an ongoing chance for an S1/Minor Storm or greater event throughout from AR3663 and AR3664.

High energy electron flux is forecast to be at background to moderate levels throughout, with associated fluence expected to remain well below the Active ($1e8$ integrated pfu) threshold.

Figure 1. Solar Analysis Valid 08/0745 UTC.



Key: Filament ____, Prominence ____, Magnetic Field Line - - -, Polarity +/-, Coronal Holes: Grey shaded area CHxx +/-, Sunspot groups 25xx - Mt Wilson α - β - $\beta\gamma$ - $\beta\gamma\delta$ and Zurich-McIntosh Axx etc.

Geomagnetic Storms:

There are multiple potential Earth-directed CMEs in the guidance period, although confidence is low, particularly on Days 1 and 2 (08 and 09 May) due to a recent lack of SOHO imagery. A mostly northbound CME in association with the X4.5 flare on 06 May has the potential to be partially Earth-directed with a slight chance of a glancing impact early on Day 2 (09 May). Limited imagery continues to give a challenging analysis of a potential second Earth-bound CME with coronal dimming in the southeast disc near AR3664 around 05/1500 UTC that led to a CME and this may arrive later on Day 3 (10 May). A 'full-halo' CME associated with the LDE X1.0 flare of AR3664 at 08/0509 UTC is still being analysed, but initial output suggests an arrival time late on Day 3 (10 May), potentially also 'sweeping up' the second CME in the forecast.

Earth remains connected to a HSS from CH35/+, with slightly elevated winds observed on a declining trend. Speeds are expected to fall back to background levels by Day 3 (10 May). No further coronal hole enhancements feature in the forecast period. The combination of declining HSS and potential CME influence through the period, as well as gaps in coronagraph imagery, results in lower than average confidence for detail of geomagnetic activity in coming days, particularly early in the period.

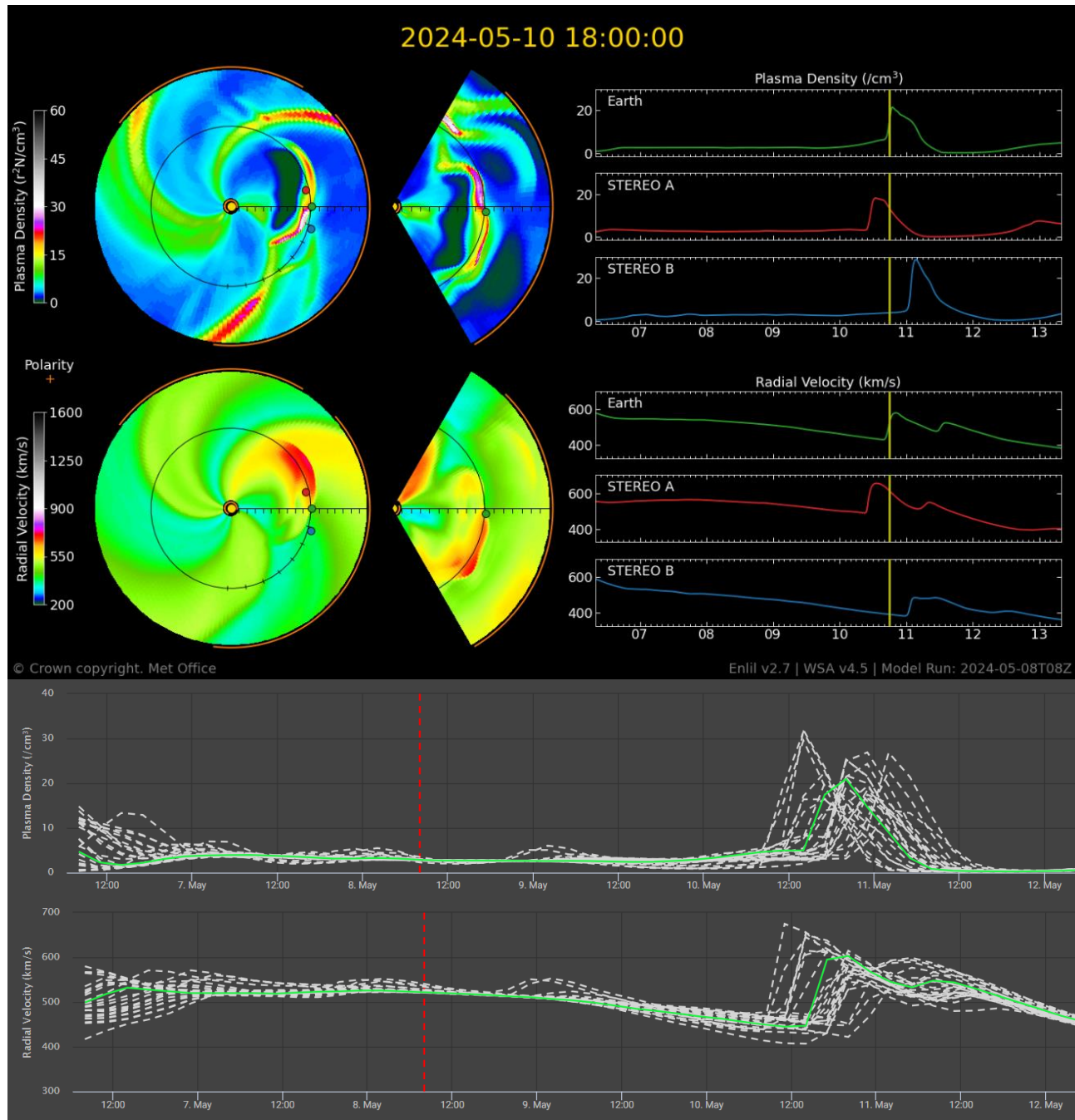
Geomagnetic activity probably mostly remaining Quiet to Unsettled through the early guidance period, with a slight chance of Minor Storm (Kp5) intervals on Days 1 and 2 (08 and 09 May). Pending the forecast 'full-halo' CME arrival on Day 3 (10 May), activity should increase to G1-G2/Minor-Moderate Storm levels and a chance of peak G3/Strong Storm conditions, with activity projected to decline into Day 4 (11 May).

Geo-Magnetic Storm	Level	Past 24 Hours (Yes/No)	Day 1 (00-24 UTC)	Day 2 (00-24 UTC)	Day 3 (00-24 UTC)	Day 4 (00-24 UTC)
Probability (Exceedance)			(%)	(%)	(%)	(%)
Minor or Moderate	G1 to G2	No	5	10	80	80
Strong	G3	No	1	1	20	20
Severe	G4	No	1	1	5	5
Extreme	G5	No	1	1	1	1

Geomagnetic Activity - Earthbound Coronal Mass Ejections

Date/time 21.5R (UTC)	Halo: Full or Partial	Source	Source Location	Estimated Speed	Estimated Arrival Time	Comments
06/1056	Partial	AR3663	NW	811 km/s	09/0000 UTC	Slight chance of glancing impact
06/0251	Partial	Filament	SE	323 km/s	10/1200 UTC	Slight chance of glancing impact
08/0927	Full	AR3664	SW	850 km/s	10/1800 UTC	CME awaiting more comprehensive analysis

Figure 2: Latest MOSWOC Enlil and Ensemble output



Radio Blackouts - X-Ray Flares:

Solar activity is High with the largest flare an X1.0 flare event from AR3663 at 08/0141 UTC. A second X1.0 flare occurred at 08/0509 UTC from AR3664. Frequent M-class flares have been observed from both regions during the past 24-hours.

Seven sunspot regions feature on the visible disc. AR3664 is the largest and most active flaring region on the disc and comprises of a number of strong spots with significant mixing of its constituent magnetic polarities. This is particularly evident in its intermediate portion and with the region displaying a complex Mt Wilson Beta-Gamma-Delta configuration containing multiple delta spots. The region lies in close proximity to AR3668 to the east, with which it is magnetically connected. AR3663 remains active with a delta spot configuration, although appears to be waning in size as it approaches the western limb. Analysis of this feature will become increasingly difficult as it approaches the northwest horizon. The remaining sunspots on the disc are smaller, more stable and magnetically simple.

AR3663 is due to rotate over the western horizon on Day 3 (10 May). Far-sided helioseismographs indicate a number of returning regions over the eastern horizon in the coming days that may supplement the present risk of significant flares should they persist.

High solar activity is expected to continue, with frequent M-class flares and a chance of isolated X-class flares throughout, principally from AR3663 and AR3664.

X Ray Flares	Level	Past 24 Hours (Yes/No)	Day 1 (00-24 UTC)	Day 2 (00-24 UTC)	Day 3 (00-24 UTC)	Day 4 (00-24 UTC)
Probability			(%)	(%)	(%)	(%)
Active	R1-R2 M Class	Yes	95	95	90	90
Very Active	R3 to R5 X Class	Yes	45	45	30	20

Figure 3a: Latest GOES16/18 X-Ray flux



Figure 3b: SDO/HMI Flat Orange (left) and Colorized Magnetogram (right) of AR3664 and AR3668 at 08/1145 UTC

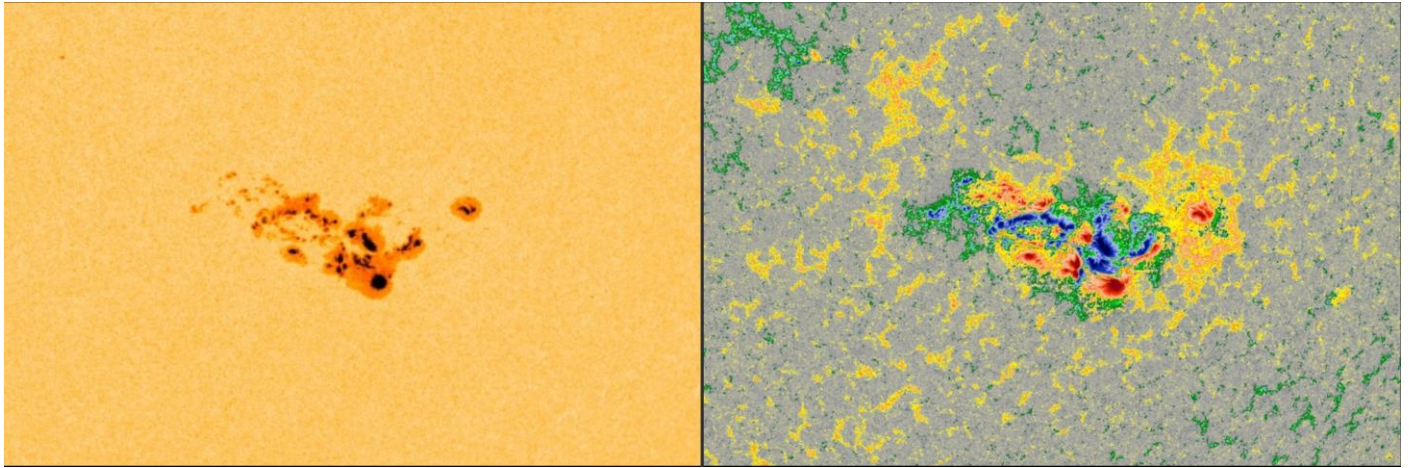
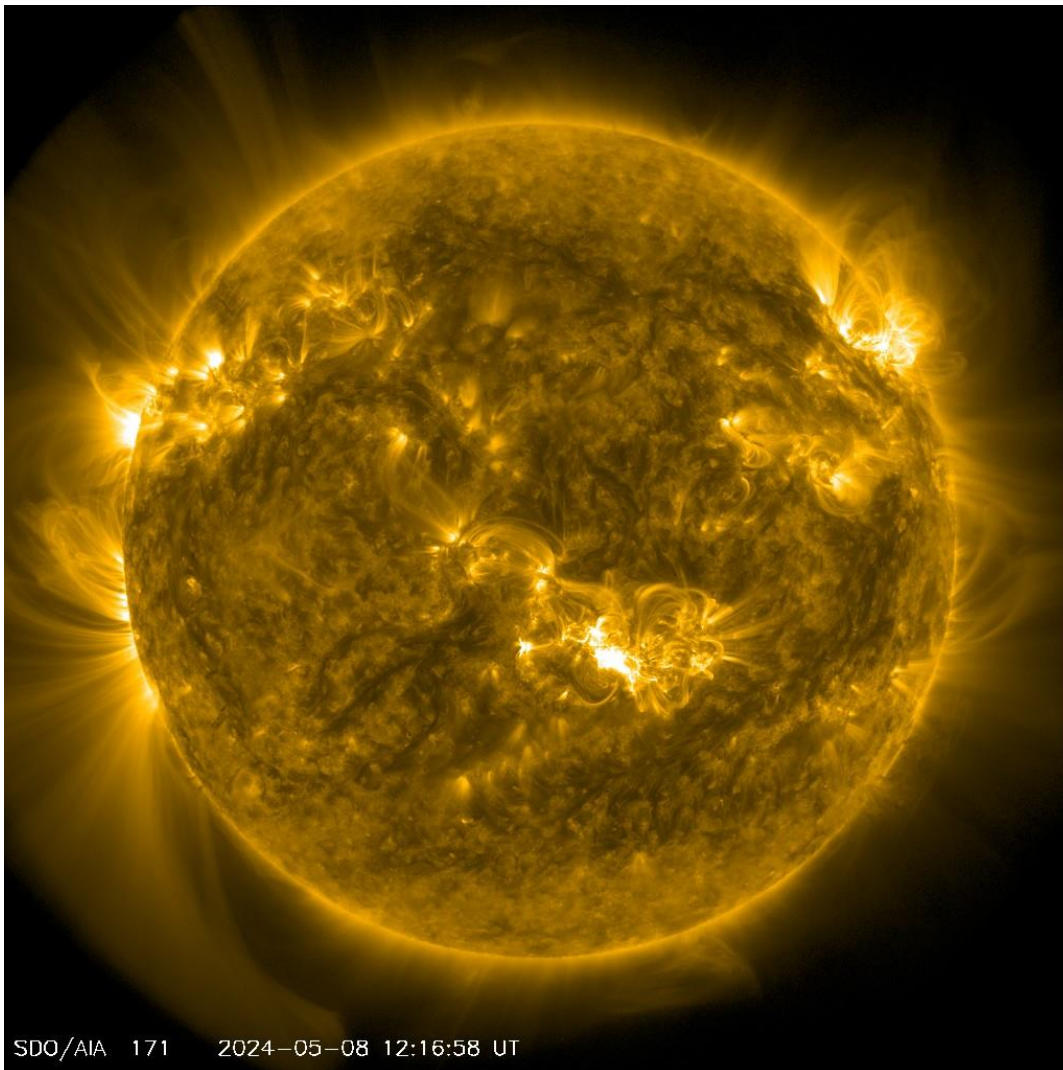


Figure 3c: SDO/AIA 171 imagery at 08/1216 UTC



Solar Radiation Storms - (High Energy Protons):

High energy proton flux (greater than 10MeV) observed at GOES-18 has been at background levels and is currently likely to remain that way through the period. However, there is a chance of an S1/Minor Storm or greater event occurring on all days, from both AR3663 and AR3664 as they transit the western solar hemisphere. Although recent flares have so far failed to trigger any significant proton response the X1.0 flare of AR3664 at 08/0509 UTC could yet illicit a more substantial increase and will be monitored closely.

Radiation Storms	Level (cm ⁻² sr ⁻¹ s ⁻¹)	Past 24 Hours (Yes/No)	Day 1 (00-24 UTC)	Day 2 (00-24 UTC)	Day 3 (00-24 UTC)	Day 4 (00-24 UTC)
Probability (Exceedance)			(%)	(%)	(%)	(%)
Active	≥ S1	No	30	30	30	25
Very Active	≥ S3 *	No	1	1	1	1
* S3 ≥ 10 MeV ≥ 1000 pfu and / or ≥ 50 MeV ≥ 10 pfu. (pfu = cm ⁻² sr ⁻¹ s ⁻¹)						

High Energy Electrons Event (≥ 2MeV):

High energy electron flux is at normal background levels. Likely CME and coronal hole influences through the period are expected to limit electron flux to background to moderate levels on all days.

Overall, the 24 hour integrated electron fluence is forecast to remain well below the Active (1e8 integrated pfu) threshold.

GEO Electron Environment	Level (cm ⁻² sr ⁻¹ day ⁻¹)	Past 24 Hours (Yes/No)	Day 1 (00-24 UTC)	Day 2 (00-24 UTC)	Day 3 (00-24 UTC)	Day 4 (00-24 UTC)
Probability (Exceedance)			(%)	(%)	(%)	(%)
Active	≥ 2 MeV ≥ 1x10 ⁸	No	1	1	1	1
Very Active	≥ 2 MeV ≥ 1x10 ⁹	No	1	1	1	1